



● Congratulations! You passed!

Grade received 100% To pass 80% or higher

Line Sweep: missing parts

Go to next item

1. Is it possible to adapt the plane sweep algorithm in the three-dimensional setting, where we use a sweep plane instead of a sweep line?
Ready to review what you've learned before taking the quiz? I'm here to help.

☐ Yes

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☒ No

🚩 Submit your assignment

Correct, defining a status structure that represents the edges hit by the sweep plane is not possible.

Try again

● Receive grade

Your grade

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2. Suppose we want to find all intersections of circles instead of line segments. If we adapted the algorithm to work with circles, how would we adapt it?
We keep your highest score.

☐ Treat a circle as two semi circles cut at the left-most and right-most point. Use a vertical sweep line and change the intersection operations to work with semi circles.

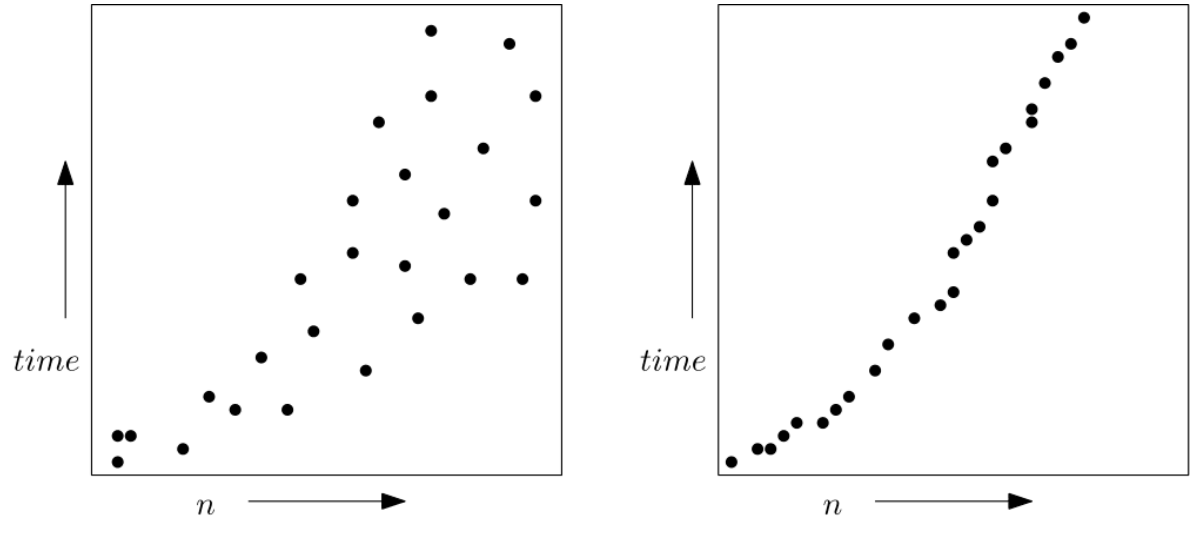
☒ Treat a circle as two semi circles cut at the top and bottom. Use a vertical sweep line and change the intersection operations to work with semi circles.

☐ Use a vertical sweep line as with line segment intersection. Change the intersection operations to work for circles.

☐ It is not possible to find circle intersections using a sweep line algorithm.

Correct

3.



1 / 1 point

Suppose we use both a brute force and a plane sweep algorithm in order to find the number of intersections in various inputs. The running times were recorded and plotted as shown in the images above. Which plot corresponds to which algorithm?

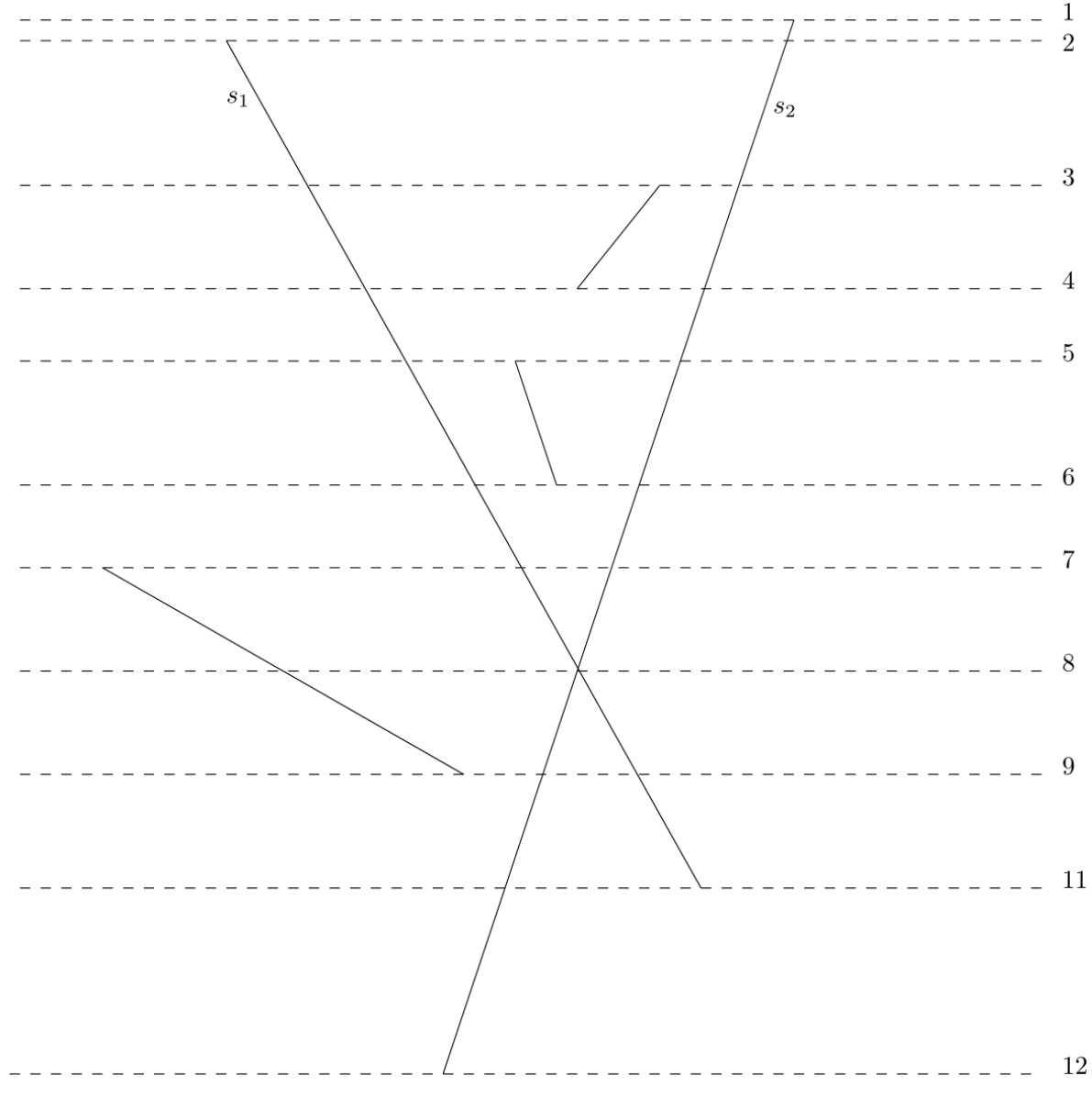
☒ Left plane sweep, right brute force.

☐ Left brute force, right plane sweep.

Correct

Correct, the plane sweep algorithm running time does not only depend on the number of points, but also on the number of intersections.

4.



1 / 1 point

In the image above a couple of line segments are shown. At each event a dashed line is shown that corresponds to the sweep line at some point in time. At which time steps is the intersection between segments **s1** and **s2** discovered? Note that it could be the case that it is discovered multiple times.

☐ 1

☒ 2

Correct

Correct, the segments become neighbors.

☐ 3

☒ 4

Correct

Correct, the segments become neighbors.

☐ 5

☒ 6

Correct

Correct, the segments become neighbors.

☐ 7

☐ 8

☐ 9

☐ 10

☐ 11

☐ 12

https://www.coursera.org/learn/geometric-algorithms/quiz/line-sweep-missing-parts/attempt1?directToCover=true

1/1